

Angel Jesús Estrada Anaya

Senior AI Automation Engineer | Senior Backend Developer

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PROFESSIONAL SUMMARY

Senior AI Automation Engineer and Backend Developer with a proven track record of architecting scalable enterprise platforms and delivering high-impact autonomous AI solutions both in corporate environments and as an independent contractor. Expert in designing production-ready multi-agent systems, advanced RAG/GraphRAG architectures, and custom Model Context Protocol (MCP) integrations using Python, n8n orchestration, and agentic frameworks like LangGraph and crewAI. Deep hands-on experience deploying conversational ecosystems (WhatsApp Official API & Telegram), real-time Voice-AI agents with LiveKit, and self-hosted analytics platforms like Metabase on dedicated server infrastructures. Highly skilled in Java (Spring Boot) and Python (FastAPI, Pandas) microservices deployed on AWS and Azure cloud environments. Adept at bridging the gap between cutting-edge AI technologies and production-grade business applications to drive measurable operational efficiency.

CORE SKILLS

- **AI Engineering:** Multi-Agent Systems, GraphRAG Systems, Multimodal AI Pipelines, Model Context Protocol (MCP) Implementations.
- **Software Architecture:** Backend Microservices, RESTful API Design, Enterprise Systems Integration, Distributed Systems.
- **Cloud & Infrastructure:** Multitenant Cloud Architecture, Infrastructure as Code (IaC), CI/CD Automation, Container Orchestration.
- **Methodologies:** Technical Leadership, Agile/Scrum Frameworks, Cross-Functional Collaboration, System Performance Optimization.

PROFESSIONAL EXPERIENCE

AI Solutions Engineer (Contractor)

Self-Employed | Remote

January 2025 – Present

An independent software engineering initiative delivering production-ready AI orchestration, workflow automation, and data analytics architectures to enhance operational efficiency and system reliability.

- **Production-Grade Chatbots & BI:** Architect multimedia AI agents on WhatsApp/Telegram using Python and n8n, integrating automated Excel data processing for scheduling and self-hosted Metabase dashboards for interactive analytics.
- **Multi-Agent Orchestration:** Design and develop production business workflows using n8n, while engineering proof-of-concept autonomous networks utilizing OpenAI Agent.
- **Advanced RAG & GraphRAG:** Implement advanced RAG and GraphRAG architectures using Python and n8n to enable deep contextual data retrieval for autonomous OpenAI Agents.
- **Next-Gen AI Infrastructure:** Build and self-host custom Model Context Protocol (MCP) integrations and real-time Voice-AI agents leveraging LiveKit, executing full server deployment for low-latency interactions.

Key Results: Deployed low-latency Voice AI | Automated end-to-end scheduling pipelines | Delivered interactive real-time BI dashboards.

Senior Backend & AI Automation Engineer (Promoted from Semi-Senior Backend Developer, Sept 2024)

Transborder | Bogotá, Colombia (Remote)

November 2023 – Present

Logistics-technology company delivering automated freight tariff processing, customs, and cross-border trade solutions.

- **AI-Powered Rate Ingestion System:** Architect and deploy an automated freight tariff processing pipeline, integrating n8n for secure customer data ingestion that triggers a high-performance Python (FastAPI) microservice to parse complex rate files with Pandas.
- **Agentic Workflows & RAG:** Implement autonomous agentic decision-making flows using LangGraph and advanced RAG architectures, leveraging Supabase for vector storage and semantic retrieval.
- **Enterprise Multi-tenant Security:** Design and secure a multitenant SaaS architecture with strict data isolation using Azure B2C, Azure App Configuration, and Azure Key Vault.
- **Backend & Cloud Reliability:** Engineer and maintain mission-critical Java 11 / Spring Boot microservices deployed on Kubernetes, handling high-load logistics data with microsecond-level latency.
- **DevOps & Infrastructure as Code:** Automate infrastructure provisioning and software deployments through secure CI/CD pipelines built with Azure DevOps and Terraform.
- **Database Optimization:** Rewrite and optimize legacy SQL Server transactions, indexes, and stored procedures to improve query performance and API response times.

- **Monitoring & Alerting:** Configure proactive monitoring dashboards using Prometheus and Grafana, enabling early incident detection across integration services.
- Key Results: 70%+ reduction in manual rate classification time | 35% increase in query performance | 97% uptime on critical integrations

Software Engineer Expert

Carvajal Tecnología y Servicios S.A.S | Bogotá, Colombia (Remote)
September 2022 – October 2023
Enterprise technology and systems-integration company delivering B2B software solutions for corporate clients across Latin America.

- **Enterprise Integrations:** Developed and maintained highly available Java 8/11 microservices using Spring Boot and Apache Camel to streamline complex enterprise B2B data exchanges.
- **Asynchronous Messaging:** Integrated and configured ActiveMQ messaging brokers, boosting system resilience and horizontal scalability under peak transactional loads.
- **Code Quality & Security Guardrails:** Enforced clean-code standards and static analysis using SonarQube, CheckStyle, and Jacoco.
- **Cloud Deployment:** Deployed scalable REST APIs on AWS EC2 and handled workload orchestration using Kubernetes auto-scaling groups.

Key Results: 85%+ unit test coverage sustained | Increased system resilience under peak transactional loads | Scalable REST APIs on auto-scaling infrastructure

EDUCATION

B.S. in Mechatronics Engineering
Universidad ECCI — Bogotá, Colombia | 2020 – 2025

Technologist in Automation and Industrial Robotics
Universidad ECCI — Bogotá, Colombia | 2020 – 2023

CERTIFICATIONS

- Multi AI Agent Systems with crewAI — DeepLearning.AI (2025)
- N8N Self-Hosted for Enterprise & Professional Workflows (3 Certifications) — Platzi (2026)
- Prompt Engineering for Developers — DeepLearning.AI (2025)
- Samsung Innovation Campus: Artificial Intelligence — Universidad del Rosario (2025)
- Specialization in DevOps — MundosE (2024 – 2025)
- Specialization in Data Science — ESAP Colombia (2022)
- Certified Tech Developer — Digital House (2021 – 2022)
- Scrum Fundamentals Certified (SFC) — SCRUMstudy (2022)

SKILLS & TOOLS

- **AI & Automation:** n8n, LangGraph, crewAI, LangChain, OpenAI API, Anthropic API, LiveKit, Supabase, Pgvector.
 - **Databases & Messaging:** SQL Server, MySQL, Oracle, MongoDB, Redis, ActiveMQ, Metabase, Webhooks.
- **Languages & Frameworks:** Python (FastAPI, Flask, Pandas), Java 8/11/17+ (Spring Boot, Spring Security), JavaScript (React.js), Apache Camel.
 - **DevOps & Cloud:** Microsoft Azure, AWS (EC2, S3), Terraform, Docker,
- Kubernetes, Jenkins, Azure DevOps, Git/GitHub.

LANGUAGES

Spanish Native	English B1 – Professional Working Proficiency
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